

Oracle stack — 10 independent code paths

8 non-cctbx · trust comes from cross-family agreement, not from any single tool

cctbx family (1)

PHENIX (mmtbx · phenix.* binaries)

non-cctbx (9 binaries across 8 code paths)

MolProbity standalone	Richardson lab — probe + reduce
TM-align	Zhang lab — sequence-independent superposition
gemmi	Global Phasing / CCP4 — sfcalc, R-factors, RMSD
Servalcat	Murshudov — sigmaa, σ_A maps
OpenStructure (OST)	SIB — IDDT, structure assessment
REFMAC5	CCP4 — independent refinement engine, NCYC=0 R-factors
ProSMART	CCP4 / Murshudov — Procrustes geometry analysis
CheckMyMetal-local	Zheng-criteria geometry classifier (local heuristic)
ctruncate (NEW)	CCP4 — Wilson B · L-test twinning · ΔB aniso · tNCS · ice rings

Cross-tool coverage by task — fully closed

Task	cctbx	non-cctbx	Status
T01 RMSD/superpose	phenix.superpose_models	TM-align · gemmi · OST · ProSMART	closed
T03 R-factors	phenix.model_vs_data	gemmi · Servalcat · REFMAC5	closed
T05 Geometry	mmtbx.validation_summary	MolProbity · ProSMART · REFMAC5	closed
T06 Density fit	phenix.real_space_correlation	gemmi · Servalcat	closed
T10 Ligand / metal	phenix.real_space_correlation	CheckMyMetal-local	closed
T13 Data quality	phenix.model_vs_data	ctruncate · CCP4 aimless	closed